

PROJECT DOCUMENT

UK Weekly Road Fuel Price Analysis

Junior Data Analyst Portfolio Project

1. Project Objective

Build a clean, well-documented Excel workbook that turns official UK weekly road-fuel price statistics into:

- A reliable, cleaned dataset ready for analysis,
- Clear weekly and monthly price insights, and
- A simple interactive lookup dashboard where a user can enter a date (or Week ID) and instantly see petrol price, diesel price, month, and week-on-week change.

The project uses only the real columns present in the government data file. No invented fields.

2. Business Questions

The analysis is designed to answer practical questions that a transport, retail or policy stakeholder might ask:

1. What is the current average UK pump price for unleaded petrol and for diesel?
2. How much did prices change compared with the previous week and the same week last year?
3. How have average monthly prices moved over the last several years?
4. In which periods did petrol or diesel reach their highest and lowest levels?
5. What is the typical diesel premium (diesel price minus petrol price) at any given week?
6. Can I quickly retrieve the exact price and change figures for any historical week?

3. Key Performance Indicators (KPIs)

All KPIs are calculated directly from the cleaned data:

- Latest Petrol Price (p/litre)
- Latest Diesel Price (p/litre)
- Petrol Week-on-Week Change (p/litre)
- Diesel Week-on-Week Change (p/litre)
- Petrol Year-on-Year Change (p/litre)
- Diesel Year-on-Year Change (p/litre)
- All-time Average / Maximum / Minimum price for each fuel
- Monthly Average Petrol & Diesel prices (via Pivot Table)
- Number of weeks of data available per month/year

4. Dataset & Column Definitions

4.1 Source

Official statistics published by the UK Department for Energy Security and Net Zero.

UK Weekly Fuel Price Analysis

File used: Weekly_Fuel_Prices_24-08-26.xlsx

Sheet used: Data

Coverage: Weekly observations from 9 June 2003 to 24 August 2026 (~1,212 rows).

Sample: Approximately 60 % of UK road-fuel market volume (six companies: four oil majors + two supermarket chains).

4.2 Original Columns (Data sheet, row 8)

Col	Original Header	Meaning
A	Date	Survey week date (usually Monday)
B	ULSP: Pump price (p/litre)	Average unleaded petrol pump price
C	ULSP: Diff on previous week	Petrol change vs prior week
D	ULSP: Diff on previous year	Petrol change vs same week prior year
E	ULSP: Duty rate (p/litre)	Fuel duty on petrol
F	ULSP: VAT rate (%)	VAT rate applied to petrol
G	ULSD: Pump price (p/litre)	Average diesel pump price
H	ULSD: Diff on previous week	Diesel change vs prior week
I	ULSD: Diff on previous year	Diesel change vs same week prior year
J	ULSD: Duty rate (p/litre)	Fuel duty on diesel
K	ULSD: VAT rate (%)	VAT rate applied to diesel

4.3 Cleaned Column Names Used in the Project

For readability the long headers were renamed (no data was altered):

- Date → Date
- ULSP: Pump price → Petrol_Price (p/litre)
- ULSP: Diff on previous week → Petrol_Week_Change (p/litre)
- ULSP: Diff on previous year → Petrol_Year_Change (p/litre)
- ULSP: Duty rate → Petrol_Duty (p/litre)
- ULSP: VAT rate → Petrol_VAT rate(%)
- ULSD: Pump price → Diesel_Price (p/litre)
- ULSD: Diff on previous week → Diesel_Week_Change (p/litre)
- ULSD: Diff on previous year → Diesel_Year_Change (p/litre)
- ULSD: Duty rate → Diesel_Duty (p/litre)
- ULSD: VAT rate → Diesel_VAT rate (%)

4.4 Helper Columns Added (derived only)

- Year =YEAR(Date)
- Month =MONTH(Date)
- MonthName =TEXT(Date,"MMMM")
- WeekID =YEAR(Date)&"-"&TEXT(WEEKNUM(Date,2),"00") (e.g. 2026-34)

These helpers enable monthly aggregation and the search dashboard. They do not invent new source data.

5. Workbook Structure

Sheet	Content
01_ReadMe	Project overview, data source, how to use the file, column glossary
02_RawData	Source Raw Data
03_CleanData	Full cleaned time series with renamed columns + Year/Month/WeekID helpers
04_Analysis & Findings	Latest-week KPIs, all-time high/low/average, monthly Pivot Table summary, trend charts
05_Dashboard	Interactive lookup: enter Date or WeekID → instant petrol & diesel price, changes, month

6. Recruiter Value

This project demonstrates that the candidate can:

- Work with real, publicly available government data (not toy datasets).
- Clean and standardise messy column names while preserving original meaning.
- Create reusable helper columns for time-based analysis.
- Build both summary analysis (KPIs + Pivot Table) and an interactive lookup tool.
- Document the work clearly so another analyst can understand and extend it.
- Deliver a complete, self-contained Excel workbook suitable for a junior analyst portfolio.

The scope is deliberately realistic for a junior role: solid fundamentals, clean structure, and one practical interactive feature — without over-engineering.

7. Skills Demonstrated

- Data import and sheet organisation
- Column renaming and data dictionary thinking
- Excel formulas: YEAR, MONTH, TEXT, WEEKNUM, AVERAGE, MAX, MIN, LOOKUP, INDEX, MATCH, IFERROR
- Pivot Tables for monthly aggregation
- Data Validation (dropdown list for dates)
- Basic charting (line charts for price trends)
- Dashboard / lookup design
- Documentation and readability (ReadMe sheet)
- Attention to data quality (blank checks, freeze panes, filters)

8. Limitations & Honest Scope Notes

- The project uses only the weekly pump-price series. It does not attempt regional breakdowns or supermarket-vs-forecourt splits (those are not in the source file).
- Duty and VAT rates are carried forward from the source; no independent tax modelling is performed.
- The search dashboard is formula-based. It is intentionally simple and does not require VBA or Power Query.

- Charts are limited to clear trend views; advanced statistical modelling is outside the beginner scope.